

# Theoretical insights into the electronic structure and NBO features driving nonlinear optical enhancement in the donor– $\pi$ –acceptor system of OPPh<sub>3</sub>py

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**Table S1:** Optimised bond lengths (Å) for OPPh<sub>3</sub>py at the DFT B3LYP-D3(BJ)/6-311++G(2df,p) level of theory.

| Bond (Å) | <sup>a</sup> Experimental |           | Optimised | Deviation<br>( Exp <sub>avg</sub> - Calc. ) |
|----------|---------------------------|-----------|-----------|---------------------------------------------|
|          | Isomer I                  | Isomer II |           |                                             |
| P1–O2    | 1.49 (17)                 | 1.48 (18) | 1.51      | 0.03                                        |
| P1–C3    | 1.80 (2)                  | 1.81 (3)  | 1.83      | 0.03                                        |
| P1–C14   | 1.80 (3)                  | 1.80 (3)  | 1.83      | 0.03                                        |
| P1–C25   | 1.81 (2)                  | 1.81 (3)  | 1.83      | 0.02                                        |
| C3–C4    | 1.36 (3)                  | 1.38 (4)  | 1.40      | 0.02                                        |
| C3–C12   | 1.38 (4)                  | 1.38 (4)  | 1.40      | 0.02                                        |
| C4–C6    | 1.38 (4)                  | 1.39 (4)  | 1.39      | 0.01                                        |
| C6–C8    | 1.36 (4)                  | 1.36 (4)  | 1.40      | 0.03                                        |
| C8–C10   | 1.34 (4)                  | 1.36 (5)  | 1.40      | 0.04                                        |
| C10–C12  | 1.37 (4)                  | 1.38 (5)  | 1.39      | 0.02                                        |
| C14–C23  | 1.39 (4)                  | 1.38 (4)  | 1.40      | 0.01                                        |
| C14–C15  | 1.39 (3)                  | 1.38 (3)  | 1.40      | 0.01                                        |
| C15–C17  | 1.37 (4)                  | 1.38 (4)  | 1.39      | 0.02                                        |
| C17–C19  | 1.37 (5)                  | 1.36 (4)  | 1.39      | 0.02                                        |
| C19–C21  | 1.36 (4)                  | 1.37 (4)  | 1.40      | 0.02                                        |
| C21–C23  | 1.39 (4)                  | 1.38 (4)  | 1.39      | 0.01                                        |
| C25–C33  | 1.39 (3)                  | 1.39 (3)  | 1.40      | 0.01                                        |
| C25–C26  | 1.42 (3)                  | 1.40 (3)  | 1.41      | 0.01                                        |
| C26–C27  | 1.39 (4)                  | 1.38 (4)  | 1.40      | 0.01                                        |
| C26–C35  | 1.52 (4)                  | 1.52 (4)  | 1.52      | 0.01                                        |
| C27–C29  | 1.37 (4)                  | 1.38 (4)  | 1.39      | 0.02                                        |
| C29–C31  | 1.36 (4)                  | 1.36 (4)  | 1.39      | 0.03                                        |
| C31–C33  | 1.38 (3)                  | 1.38 (4)  | 1.39      | 0.02                                        |
| C35–N38  | 1.43 (3)                  | 1.44 (3)  | 1.46      | 0.03                                        |
| N38–C40  | 1.35 (4)                  | 1.36 (3)  | 1.37      | 0.02                                        |
| C40–N41  | 1.34 (3)                  | 1.33 (3)  | 1.34      | 0.01                                        |
| C40–C48  | 1.39 (4)                  | 1.40 (4)  | 1.42      | 0.02                                        |
| N41–C42  | 1.34 (4)                  | 1.35 (4)  | 1.34      | 0.01                                        |
| C42–C44  | 1.35 (5)                  | 1.35 (5)  | 1.39      | 0.04                                        |
| C44–C46  | 1.37 (6)                  | 1.37 (5)  | 1.40      | 0.03                                        |
| C46–C48  | 1.36 (5)                  | 1.35 (4)  | 1.38      | 0.03                                        |

<sup>a</sup>Numbers in parentheses represent estimated standard uncertainties in the last digit(s) of the value.

**Table S2:** Angles (°) for OPPh<sub>3</sub>py at the DFT B3LYP-D3(BJ)/6-311++G(2df,p)) level of theory.

| Angle (°)   | <sup>a</sup> Experimental |             | Optimised | Deviation<br>( Exp <sub>avg</sub> - Calc. ) |
|-------------|---------------------------|-------------|-----------|---------------------------------------------|
|             | Isomer I                  | Isomer II   |           |                                             |
| O2-P1-C3    | 112.39 (11)               | 112.62 (12) | 112.33    | 0.16                                        |
| O2-P1-C14   | 110.16 (11)               | 111.44 (12) | 111.29    | 0.79                                        |
| C3-P1-C14   | 106.34 (12)               | 106.42 (12) | 106.04    | 0.34                                        |
| O2-P1-C25   | 112.55 (11)               | 114.12 (11) | 114.13    | 1.73                                        |
| C3-P1-C25   | 106.98 (11)               | 104.94 (11) | 105.43    | 1.23                                        |
| C14-P1-C25  | 108.14 (12)               | 106.73 (12) | 107.07    | 0.99                                        |
| C4-C3-C12   | 117.70 (20)               | 118.00 (30) | 119.88    | 1.99                                        |
| C4-C3-P1    | 124.10 (20)               | 118.30 (20) | 122.51    | 1.90                                        |
| C12-C3-P1   | 118.00 (20)               | 123.60 (20) | 117.61    | 2.00                                        |
| C3-C4-C6    | 120.90 (20)               | 120.50 (30) | 119.97    | 0.83                                        |
| C8-C6-C4    | 119.80 (30)               | 120.50 (30) | 120.02    | 0.36                                        |
| C10-C8-C6   | 120.20 (30)               | 119.70 (30) | 120.09    | 0.11                                        |
| C8-C10-C12  | 120.20 (30)               | 120.50 (30) | 120.09    | 0.31                                        |
| C10-C12-C3  | 121.20 (30)               | 120.80 (30) | 119.94    | 0.93                                        |
| C23-C14-C15 | 118.80 (30)               | 118.40 (30) | 119.83    | 0.81                                        |
| C23-C14-P1  | 123.30 (20)               | 119.40 (20) | 123.08    | 1.23                                        |
| C15-C14-P1  | 117.90 (20)               | 122.30 (20) | 117.09    | 1.06                                        |
| C17-C15-C14 | 120.40 (30)               | 120.30 (30) | 119.97    | 0.33                                        |
| C15-C17-C19 | 120.30 (30)               | 120.20 (30) | 120.09    | 0.21                                        |
| C21-C19-C17 | 120.20 (30)               | 120.50 (30) | 120.06    | 0.27                                        |
| C19-C21-C23 | 120.30 (30)               | 119.40 (30) | 120.05    | 0.17                                        |
| C21-C23-C14 | 120.00 (30)               | 121.20 (30) | 120.00    | 0.20                                        |
| C33-C25-C26 | 118.80 (20)               | 119.20 (20) | 119.85    | 0.77                                        |
| C33-C25-P1  | 120.60 (20)               | 120.30 (19) | 120.19    | 0.36                                        |
| C26-C25-P1  | 120.59 (19)               | 120.45 (19) | 119.96    | 0.54                                        |
| C27-C26-C25 | 118.00 (20)               | 118.20 (20) | 118.46    | 0.33                                        |
| C27-C26-C35 | 120.20 (20)               | 121.40 (20) | 118.82    | 1.39                                        |
| C25-C26-C35 | 121.70 (20)               | 120.40 (20) | 122.66    | 1.38                                        |
| C29-C27-C26 | 121.50 (30)               | 121.80 (30) | 121.35    | 0.28                                        |
| C31-C29-C27 | 120.80 (30)               | 120.00 (30) | 120.01    | 0.39                                        |
| C29-C31-C33 | 119.10 (30)               | 119.40 (30) | 119.60    | 0.50                                        |
| C31-C33-C25 | 121.70 (30)               | 121.40 (30) | 120.73    | 0.83                                        |

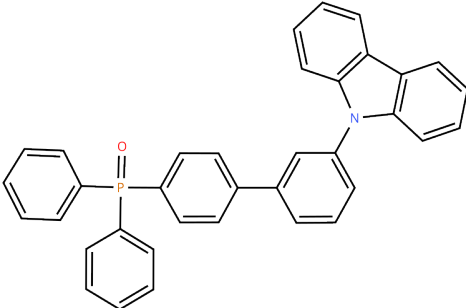
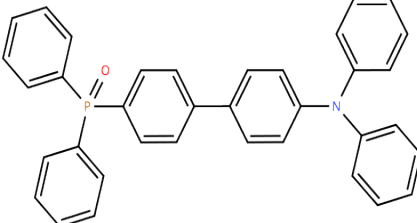
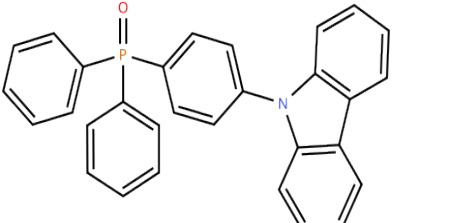
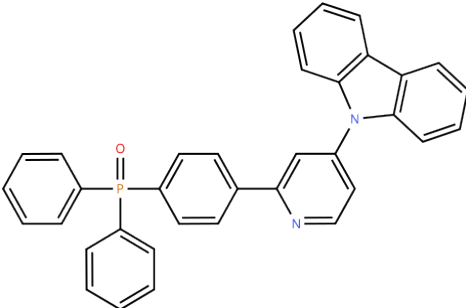
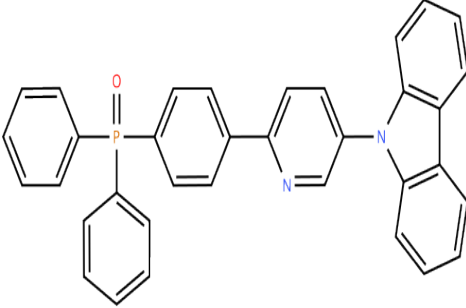
**Table S2:** (Continue)

| Angle (°)   | <sup>a</sup> Experimental |             | Optimised | Deviation<br>( Exp <sub>avg</sub> - Calc. ) |
|-------------|---------------------------|-------------|-----------|---------------------------------------------|
|             | Isomer I                  | Isomer II   |           |                                             |
| N38–C35–C26 | 115.50 (20)               | 115.40 (20) | 113.20    | 2.15                                        |
| C40–N38–C35 | 122.20 (30)               | 122.90 (20) | 121.75    | 0.83                                        |
| N41–C40–N38 | 117.70 (30)               | 118.10 (30) | 118.29    | 0.39                                        |
| N41–C40–C48 | 121.60 (30)               | 122.90 (30) | 121.99    | 0.19                                        |
| N38–C40–C48 | 120.70 (20)               | 119.10 (20) | 119.71    | 1.00                                        |
| C40–N41–C42 | 117.30 (30)               | 115.90 (30) | 117.99    | 0.55                                        |
| N41–C42–C44 | 124.80 (40)               | 124.60 (30) | 124.34    | 0.41                                        |
| C42–C44–C46 | 117.40 (40)               | 118.40 (30) | 117.40    | 0.00                                        |
| C48–C46–C44 | 120.60 (40)               | 119.50 (30) | 119.57    | 1.03                                        |
| C46–C48–C40 | 118.40 (40)               | 118.80 (30) | 118.70    | 0.25                                        |

<sup>a</sup> Numbers in parentheses represent estimated standard uncertainties in the last digit(s) of the value.

The following molecules ([Table S3](#)) were selected to benchmark the global reactivity parameters of OPPh<sub>3</sub>py. These molecules were chosen because of their similarity in structural formula, each having triphenylphosphine oxide as the base structural unit. Although the reference molecules selected share a common structural framework with OPPh<sub>3</sub>py, their reported reactivity parameters were not obtained at the same level of theory used in our calculations (B3LYP-D3(BJ)/6-311G++(2df,p)). As a result, the comparison presented here should be regarded as qualitative rather than quantitative. Nonetheless, structural similarity among the chosen molecules provides a reasonable basis for establishing general trends. To strengthen the validity of our analysis, all comparisons are interpreted cautiously, and the limitations arising from cross-method data are explicitly acknowledged.

**Table S3:** Molecular structures of the reference compounds used for benchmarking the global reactivity parameters of  $\text{OPPh}_3\text{py}$ .

| Molecule                             | Rename | Structural Formula                                                                   |
|--------------------------------------|--------|--------------------------------------------------------------------------------------|
| <b>1</b> Ref. <sup>1</sup>           | TPP1   |    |
| <b>POA</b> Ref. <sup>2</sup>         | TPP2   |    |
| <b>MPO12</b> Ref. <sup>3</sup>       | TPP3   |   |
| <b>pPO-ppy-mCZ</b> Ref. <sup>4</sup> | TPP4   |  |
| <b>pPO-ppy-pCZ</b> Ref. <sup>4</sup> | TPP5   |  |

**Table S4:** Local reactivity parameters: Fukui functions using CM5 population analysis: local softness ( $s_k^\pm$ ) and philicity ( $\omega_k^\pm$ ) in eV for selected atomic sites of OPPh<sub>3</sub>py. Values are scaled by  $10^{-2}$  for  $f_k^\pm$  and  $s_k^\pm$ , and by  $10^{-1}$  for  $\omega_k^\pm$ .

| Atom | Fukui function |         | Local softness |         | Philicity    |              |
|------|----------------|---------|----------------|---------|--------------|--------------|
|      | $f_k^+$        | $f_k^-$ | $s_k^+$        | $s_k^-$ | $\omega_k^+$ | $\omega_k^-$ |
| P1   | 3.64           | 0.09    | 1.04           | 0.03    | 6.71         | 0.17         |
| O2   | 2.37           | 0.01    | 0.68           | 0.00    | 4.37         | 0.02         |
| C3   | 2.53           | -0.70   | 0.72           | -0.20   | 4.67         | -1.29        |
| C4   | 3.08           | -0.13   | 0.88           | -0.04   | 5.68         | -0.24        |
| C6   | 2.71           | 0.90    | 0.78           | 0.26    | 5.00         | 1.66         |
| C8   | 5.58           | 1.46    | 1.60           | 0.42    | 10.29        | 2.69         |
| C10  | 4.00           | 0.98    | 1.15           | 0.28    | 7.38         | 1.80         |
| C12  | 2.28           | -0.07   | 0.65           | -0.02   | 4.20         | -0.13        |
| C14  | 2.61           | -0.54   | 0.75           | -0.15   | 4.81         | -0.99        |
| C15  | 3.60           | -0.18   | 1.03           | -0.05   | 6.64         | -0.32        |
| C17  | 3.08           | 0.83    | 0.88           | 0.24    | 5.68         | 1.53         |
| C19  | 5.74           | 1.41    | 1.64           | 0.40    | 10.59        | 2.60         |
| C21  | 3.64           | 1.06    | 1.04           | 0.30    | 6.71         | 1.96         |
| C23  | 1.87           | 0.21    | 0.54           | 0.06    | 3.45         | 0.39         |
| C25  | 1.73           | 0.54    | 0.50           | 0.15    | 3.19         | 0.99         |
| C26  | 1.61           | -0.66   | 0.46           | -0.19   | 2.97         | -1.22        |
| C27  | 2.13           | 0.33    | 0.61           | 0.09    | 3.93         | 0.61         |
| C29  | 4.27           | 1.55    | 1.22           | 0.44    | 7.87         | 2.86         |
| C31  | 2.47           | 2.76    | 0.71           | 0.79    | 4.56         | 5.09         |
| C33  | 1.39           | 1.72    | 0.40           | 0.49    | 2.56         | 3.17         |
| C35  | 0.45           | 2.16    | 0.13           | 0.62    | 0.83         | 3.98         |
| N38  | -0.17          | 15.34   | -0.05          | 4.39    | -0.31        | 28.29        |
| C40  | -0.06          | 3.34    | -0.02          | 0.96    | -0.11        | 6.16         |
| N41  | 0.23           | 5.75    | 0.07           | 1.65    | 0.43         | 10.60        |
| C42  | 1.28           | 6.09    | 0.37           | 1.74    | 2.36         | 11.23        |
| C44  | 1.78           | 10.46   | 0.51           | 2.99    | 3.29         | 19.29        |
| C46  | 1.78           | 4.53    | 0.51           | 1.30    | 3.29         | 8.36         |
| C48  | 0.64           | 7.76    | 0.18           | 2.22    | 1.18         | 14.32        |

**Table S5:** Second order perturbation theory analysis of the Fock matrix in the NBO basis for OPPh<sub>3</sub>py.

| Donor ( <i>i</i> ) | Type | ED( <i>i</i> )<br>(e) | Acceptor ( <i>j</i> ) | ED( <i>j</i> )<br>(e) | Type | <sup>a</sup> <i>E</i> (2)<br>(kcal/mol) | <sup>b</sup> <i>E</i> ( <i>j</i> ) − <i>E</i> ( <i>i</i> )<br>(a.u) | <sup>c</sup> <i>F</i> ( <i>i</i> , <i>j</i> )<br>(a.u) |
|--------------------|------|-----------------------|-----------------------|-----------------------|------|-----------------------------------------|---------------------------------------------------------------------|--------------------------------------------------------|
| LP O2              | -    | 1.976                 | BD* N38-H39           | 0.028                 | σ*   | 2.94                                    | 1.03                                                                | 0.049                                                  |
| LP O2              | -    | 1.976                 | RY P1                 | 0.009                 | -    | 7.03                                    | 1.76                                                                | 0.099                                                  |
| LP O2              | -    | 1.833                 | BD* P1-C14            | 0.118                 | π*   | 17.75                                   | 0.5                                                                 | 0.084                                                  |
| LP O2              | -    | 1.833                 | RY P1                 | 0.029                 | -    | 8.36                                    | 1.26                                                                | 0.092                                                  |
| LP O2              | -    | 1.832                 | BD* P1-C3             | 0.125                 | π*   | 17.96                                   | 0.49                                                                | 0.084                                                  |
| LP O2              | -    | 1.832                 | BD* P1-C25            | 0.123                 | π*   | 9.49                                    | 0.49                                                                | 0.061                                                  |
| LP O2              | -    | 1.832                 | RY* P1                | 0.030                 | -    | 8.54                                    | 1.23                                                                | 0.091                                                  |
| LP N38             | -    | 1.743                 | BD* C26-C35           | 0.035                 | π*   | 8.2                                     | 0.64                                                                | 0.065                                                  |
| LP N38             | -    | 1.743                 | BD* C40-N41           | 0.460                 | π*   | 53.84                                   | 0.26                                                                | 0.105                                                  |
| LP N41             | -    | 1.909                 | BD* C40-C48           | 0.032                 | π*   | 9.68                                    | 0.83                                                                | 0.08                                                   |
| LP N41             | -    | 1.909                 | BD* C42-C44           | 0.243                 | π*   | 8.53                                    | 0.87                                                                | 0.077                                                  |
| BD C3-C4           | π    | 1.681                 | BD* C6-C8             | 0.322                 | π*   | 18.13                                   | 0.29                                                                | 0.065                                                  |
| BD C3-C12          | π    | 1.681                 | BD* C10-C12           | 0.303                 | π*   | 20.33                                   | 0.29                                                                | 0.069                                                  |
| BD C6-C8           | π    | 1.650                 | BD* C3-C4             | 0.377                 | π*   | 23.02                                   | 0.27                                                                | 0.071                                                  |
| BD C6-C8           | π    | 1.650                 | BD* C10-C12           | 0.303                 | π*   | 19.32                                   | 0.28                                                                | 0.066                                                  |
| BD C10-C12         | π    | 1.642                 | BD* C3-C4             | 0.377                 | π*   | 20.34                                   | 0.27                                                                | 0.066                                                  |
| BD C10-C12         | π    | 1.642                 | BD* C6-C8             | 0.322                 | π*   | 21.79                                   | 0.28                                                                | 0.069                                                  |
| BD C14-C15         | π    | 1.671                 | BD* C17-C19           | 0.319                 | π*   | 17.98                                   | 0.29                                                                | 0.064                                                  |
| BD C14-C15         | π    | 1.671                 | BD* C21-C23           | 0.315                 | π*   | 21.37                                   | 0.28                                                                | 0.07                                                   |
| BD C17-C19         | π    | 1.650                 | BD* C14-C15           | 0.360                 | π*   | 23.08                                   | 0.28                                                                | 0.071                                                  |
| BD C17-C19         | π    | 1.650                 | BD* C21-C23           | 0.315                 | π*   | 19.85                                   | 0.28                                                                | 0.066                                                  |
| BD C21-C23         | π    | 1.657                 | BD* C14-C15           | 0.360                 | π*   | 19.27                                   | 0.28                                                                | 0.065                                                  |
| BD C21-C23         | π    | 1.657                 | BD* C17-C19           | 0.319                 | π*   | 20.91                                   | 0.28                                                                | 0.069                                                  |
| BD C25-C33         | π    | 1.694                 | BD* C26-C27           | 0.305                 | π*   | 19.88                                   | 0.3                                                                 | 0.069                                                  |
| BD C25-C33         | π    | 1.694                 | BD* C29-C31           | 0.320                 | π*   | 17.25                                   | 0.29                                                                | 0.063                                                  |
| BD C26-C27         | π    | 1.629                 | BD* C25-C33           | 0.373                 | π*   | 20.03                                   | 0.26                                                                | 0.065                                                  |
| BD C26-C27         | π    | 1.629                 | BD* C29-C31           | 0.320                 | π*   | 22.31                                   | 0.27                                                                | 0.07                                                   |
| BD C29-C31         | π    | 1.656                 | BD* C25-C33           | 0.373                 | π*   | 23.6                                    | 0.27                                                                | 0.072                                                  |
| BD C29-C31         | π    | 1.656                 | BD* C26-C27           | 0.305                 | π*   | 18.8                                    | 0.29                                                                | 0.066                                                  |
| BD C40-N41         | π    | 1.710                 | BD* C42-C44           | 0.311                 | π*   | 30.15                                   | 0.32                                                                | 0.088                                                  |
| BD C42-C44         | π    | 1.697                 | BD* C40-N41           | 0.460                 | π*   | 13.34                                   | 0.26                                                                | 0.053                                                  |
| BD C42-C44         | π    | 1.697                 | BD* C46-C48           | 0.288                 | π*   | 24.22                                   | 0.28                                                                | 0.074                                                  |
| BD C46-C48         | π    | 1.732                 | BD* C40-N41           | 0.460                 | π*   | 28.86                                   | 0.27                                                                | 0.079                                                  |
| BD C46-C48         | π    | 1.732                 | BD* C42-C44           | 0.311                 | π*   | 14.35                                   | 0.29                                                                | 0.057                                                  |

<sup>a</sup> *E*(2) means energy of hyper conjugative interactions. <sup>b</sup> Energy difference between donor and acceptor *i* and *j* NBO orbitals

<sup>c</sup> *F*(*i*, *j*) is the Fock matrix element between *i* and *j* NBO orbitals



Nonlinear optical (NLO) properties reported in the following supporting tables were calculated using the levels of theory summarised below:

| Method         | Empirical Dispersion | Basis Set       |
|----------------|----------------------|-----------------|
| B3LYP          | GD3(BJ)              | 6-311++G(2df,p) |
| CAM-B3LYP      | GD3(BJ)              | 6-311++G(2df,p) |
| M06-2X         | GD3                  | 6-311++G(2df,p) |
| *M06-2X        | GD3                  | def2-TZVPPD     |
| $\omega$ B97XD | –                    | 6-311++G(2df,p) |

All properties were calculated using different solvation models: gas phase (GAS), polarisable continuum model (PCM) and solvation model based on density(SMD) and at multiple wavelengths ( $\lambda = \infty$ , 532, 625, 1064, 1341, and 1906 nm). However, values corresponding to 532, and 625 nm are not reported in the main supporting tables due to the large discrepancies and irregular behaviour observed in these data. The complete raw dataset is available at: <https://doi.org/10.5281/zenodo.18866407>.

**Table S6:** Calculated dipole moment ( $\mu_D$ ), isotropic polarisability ( $\alpha_{\text{iso}} \times 10^{-24}$  esu), and anisotropic polarisability ( $\alpha_{\text{aniso}} \times 10^{-24}$  esu) of OPPh<sub>3</sub>py at different wavelengths and levels of theory.

| Model | $\lambda$ | $\mu_D$ |           |        |         |                | $\alpha_{\text{iso}}$ |           |        |         |                | $\alpha_{\text{aniso}}$ |           |        |         |                |
|-------|-----------|---------|-----------|--------|---------|----------------|-----------------------|-----------|--------|---------|----------------|-------------------------|-----------|--------|---------|----------------|
|       |           | B3LYP   | CAM-B3LYP | M06-2X | *M06-2X | $\omega$ B97XD | B3LYP                 | CAM-B3LYP | M06-2X | *M06-2X | $\omega$ B97XD | B3LYP                   | CAM-B3LYP | M06-2X | *M06-2X | $\omega$ B97XD |
| GAS   | 1064      | 4.80    | 4.76      | 4.57   | 4.58    | 4.67           | 48.25                 | 46.84     | 46.83  | 47.31   | 47.01          | 11.26                   | 10.54     | 10.74  | 10.85   | 10.63          |
|       | 1341      | 4.80    | 4.76      | 4.57   | 4.58    | 4.67           | 47.97                 | 46.59     | 46.58  | 47.07   | 46.76          | 11.11                   | 10.41     | 10.61  | 10.72   | 10.50          |
|       | 1906      | 4.80    | 4.76      | 4.57   | 4.58    | 4.67           | 47.73                 | 46.38     | 46.37  | 46.86   | 46.55          | 10.98                   | 10.31     | 10.50  | 10.61   | 10.40          |
|       | $\infty$  | 4.80    | 4.76      | 4.57   | 4.58    | 4.67           | 47.51                 | 46.18     | 46.17  | 46.65   | 46.34          | 10.86                   | 10.21     | 10.39  | 10.51   | 10.29          |
| PCM   | 1064      | 6.13    | 6.07      | 5.80   | 5.83    | 5.94           | 54.82                 | 53.10     | 53.11  | 53.74   | 53.34          | 11.63                   | 10.85     | 11.02  | 11.14   | 10.95          |
|       | 1341      | 6.13    | 6.07      | 5.80   | 5.83    | 5.94           | 54.47                 | 52.79     | 52.80  | 53.42   | 53.02          | 11.45                   | 10.71     | 10.88  | 10.99   | 10.80          |
|       | 1906      | 6.13    | 6.07      | 5.80   | 5.83    | 5.94           | 54.17                 | 52.52     | 52.53  | 53.15   | 52.75          | 11.31                   | 10.59     | 10.75  | 10.86   | 10.68          |
|       | $\infty$  | 6.13    | 6.07      | 5.80   | 5.83    | 5.94           | 59.45                 | 57.54     | 57.57  | 58.35   | 57.85          | 10.42                   | 9.76      | 9.88   | 9.97    | 9.81           |
| SMD   | 1064      | 6.33    | 6.27      | 5.99   | 6.04    | 6.13           | 56.23                 | 54.46     | 54.49  | 55.14   | 54.71          | 11.75                   | 10.94     | 11.35  | 11.49   | 10.98          |
|       | 1341      | 6.33    | 6.27      | 5.99   | 6.04    | 6.13           | 55.86                 | 54.13     | 54.16  | 54.81   | 54.38          | 11.57                   | 10.79     | 11.19  | 11.34   | 10.83          |
|       | 1906      | 6.33    | 6.27      | 5.99   | 6.04    | 6.13           | 55.54                 | 53.85     | 53.88  | 54.53   | 54.10          | 11.42                   | 10.67     | 11.06  | 11.21   | 10.70          |
|       | $\infty$  | 6.33    | 6.27      | 5.99   | 6.04    | 6.13           | 62.19                 | 60.18     | 60.26  | 61.11   | 60.55          | 10.55                   | 9.84      | 10.22  | 10.35   | 9.84           |

\*Calculations for this functional were performed using the def2-TZVPPD basis set.

**Table S7:** Selected components contributing tensors and values of the static and frequency dependant first-order hyperpolarisability,  $\beta$  (static ( $\beta(0; 0, 0) \times 10^{-30}$  esu, dynamic ( $\beta^\lambda(-\omega; \omega, 0) \times 10^{-30}$  esu) of OPPh<sub>3</sub>py calculated at different levels of theory. All calculations employed the 6-311++G(2df,p) basis set unless otherwise indicated.

| Model | Method                 | $\lambda$ (nm) | $\beta_{xxx}$ | $\beta_{yyy}$ | $\beta_{zzz}$ | $\beta_x$ | $\beta_y$ | $\beta_z$ | $\beta_{  }$ | $\beta_{tot}$ |
|-------|------------------------|----------------|---------------|---------------|---------------|-----------|-----------|-----------|--------------|---------------|
| GAS   | B3LYP                  | 1064           | 0.53          | -0.47         | 1.39          | 4.66      | 7.79      | 3.01      | 1.91         | 9.56          |
|       |                        | 1341           | 0.53          | -0.46         | 1.21          | 4.57      | 7.59      | 2.51      | 1.84         | 9.21          |
|       |                        | 1906           | 0.52          | -0.46         | 1.08          | 4.50      | 7.43      | 2.14      | 1.79         | 8.94          |
|       |                        | $\infty$       | 0.35          | -0.73         | -2.77         | 5.03      | -5.59     | -4.39     | 1.74         | 8.71          |
|       | CAM-B3LYP              | 1064           | 0.71          | -0.47         | -1.03         | 5.21      | 7.32      | -5.04     | 2.06         | 10.30         |
|       |                        | 1341           | 0.70          | -0.46         | -1.01         | 5.07      | 7.14      | -4.93     | 2.01         | 10.05         |
|       |                        | 1906           | 0.68          | -0.46         | -1.00         | 4.96      | 6.99      | -4.83     | 1.97         | 9.84          |
|       |                        | $\infty$       | 0.45          | -1.08         | -1.29         | 4.21      | -8.53     | 1.49      | 1.93         | 9.63          |
|       | M06-2X                 | 1064           | 0.62          | -0.40         | -0.65         | 4.97      | 7.53      | -4.44     | 2.01         | 10.06         |
|       |                        | 1341           | 0.61          | -0.40         | -0.64         | 4.83      | 7.34      | -4.34     | 1.96         | 9.80          |
|       |                        | 1906           | 0.59          | -0.40         | -0.64         | 4.71      | 7.19      | -4.25     | 1.92         | 9.59          |
|       |                        | $\infty$       | 0.34          | -1.12         | -1.35         | 4.02      | -8.41     | 1.16      | 1.88         | 9.39          |
|       | M06-2X/<br>def2-TZVPPD | 1064           | 0.55          | -0.36         | -0.59         | 5.03      | 6.96      | -4.37     | 1.93         | 9.63          |
|       |                        | 1341           | 0.54          | -0.35         | -0.59         | 4.90      | 6.80      | -4.29     | 1.88         | 9.42          |
|       |                        | 1906           | 0.53          | -0.35         | -0.59         | 4.80      | 6.67      | -4.21     | 1.85         | 9.23          |
|       |                        | $\infty$       | 0.28          | -1.19         | -1.26         | 4.11      | -7.97     | 1.29      | 1.81         | 9.06          |
|       | $\omega$ B97XD         | 1064           | 0.61          | -0.47         | -0.96         | 4.67      | 7.30      | -5.09     | 2.01         | 10.05         |
|       |                        | 1341           | 0.60          | -0.46         | -0.94         | 4.53      | 7.12      | -4.97     | 1.96         | 9.80          |
|       |                        | 1906           | 0.58          | -0.46         | -0.93         | 4.42      | 6.98      | -4.87     | 1.92         | 9.59          |
|       |                        | $\infty$       | 0.34          | -1.13         | -1.30         | 3.66      | -8.52     | 1.50      | 1.88         | 9.40          |
| PCM   | B3LYP                  | 1064           | 0.97          | -1.07         | 0.64          | 7.64      | 11.56     | -2.97     | 2.83         | 14.17         |
|       |                        | 1341           | 0.95          | -1.06         | 0.50          | 7.43      | 11.22     | -3.22     | 2.77         | 13.84         |
|       |                        | 1906           | 0.93          | -1.04         | 0.40          | 7.26      | 10.95     | -3.41     | 2.71         | 13.57         |
|       |                        | $\infty$       | 0.50          | -1.54         | -3.70         | 7.94      | -13.12    | -1.16     | 3.08         | 15.38         |
|       | CAM-B3LYP              | 1064           | 1.18          | -1.02         | -1.73         | 8.15      | 10.62     | -10.42    | 3.39         | 16.97         |
|       |                        | 1341           | 1.15          | -1.01         | -1.68         | 7.91      | 10.33     | -10.13    | 3.30         | 16.49         |
|       |                        | 1906           | 1.13          | -1.00         | -1.65         | 7.70      | 10.10     | -9.88     | 3.22         | 16.09         |
|       |                        | $\infty$       | 0.70          | -1.78         | -1.78         | 6.77      | -15.40    | 5.59      | 3.55         | 17.73         |
|       | M06-2X                 | 1064           | 1.07          | -0.84         | -1.06         | 7.82      | 11.19     | -9.36     | 3.31         | 16.55         |
|       |                        | 1341           | 1.04          | -0.84         | -1.04         | 7.57      | 10.88     | -9.10     | 3.22         | 16.08         |
|       |                        | 1906           | 1.02          | -0.83         | -1.02         | 7.37      | 10.63     | -8.88     | 3.14         | 15.69         |
|       |                        | $\infty$       | 0.50          | -1.92         | -1.94         | 6.44      | -15.28    | 4.97      | 3.46         | 17.31         |
|       | M06-2X/<br>def2-TZVPPD | 1064           | 1.05          | -0.77         | -1.04         | 8.03      | 10.64     | -9.57     | 3.28         | 16.41         |
|       |                        | 1341           | 1.02          | -0.76         | -1.03         | 7.80      | 10.37     | -9.32     | 3.19         | 15.97         |
|       |                        | 1906           | 1.00          | -0.75         | -1.01         | 7.60      | 10.14     | -9.11     | 3.12         | 15.61         |
|       |                        | $\infty$       | 0.46          | -2.06         | -1.79         | 6.71      | -15.10    | 5.41      | 3.48         | 17.39         |
|       | $\omega$ B97XD         | 1064           | 1.15          | -1.02         | -1.60         | 7.77      | 10.64     | -10.39    | 3.36         | 16.78         |
|       |                        | 1341           | 1.12          | -1.01         | -1.56         | 7.53      | 10.36     | -10.10    | 3.26         | 16.31         |
|       |                        | 1906           | 1.10          | -0.99         | -1.53         | 7.33      | 10.14     | -9.86     | 3.19         | 15.93         |
|       |                        | $\infty$       | 0.64          | -1.87         | -1.82         | 6.30      | -15.40    | 5.62      | 3.51         | 17.57         |

**Table S7:** (Continue)

| Model | Method                 | $\lambda$ (nm) | $\beta_{xxx}$ | $\beta_{yyy}$ | $\beta_{zzz}$ | $\beta_x$ | $\beta_y$ | $\beta_z$ | $\beta_{  }$ | $\beta_{\text{tot}}$ |
|-------|------------------------|----------------|---------------|---------------|---------------|-----------|-----------|-----------|--------------|----------------------|
| SMD   | B3LYP                  | 1064           | 1.15          | -1.43         | -0.26         | 8.79      | 11.43     | -5.68     | 3.10         | 15.50                |
|       |                        | 1341           | 1.12          | -1.41         | -0.36         | 8.54      | 11.08     | -5.82     | 3.03         | 15.16                |
|       |                        | 1906           | 1.10          | -1.39         | -0.44         | 8.34      | 10.80     | -5.92     | 2.97         | 14.87                |
|       |                        | $\infty$       | 0.61          | -1.38         | -3.43         | 8.91      | -14.52    | 1.64      | 3.42         | 17.12                |
|       | CAM-B3LYP              | 1064           | 1.32          | -1.38         | -2.52         | 9.07      | 10.21     | -12.74    | 3.73         | 18.67                |
|       |                        | 1341           | 1.28          | -1.35         | -2.45         | 8.79      | 9.93      | -12.37    | 3.63         | 18.13                |
|       |                        | 1906           | 1.26          | -1.34         | -2.40         | 8.56      | 9.70      | -12.06    | 3.54         | 17.69                |
|       |                        | $\infty$       | 0.79          | -1.61         | -1.50         | 7.56      | -16.49    | 8.16      | 3.98         | 19.89                |
|       | M06-2X                 | 1064           | 1.22          | -1.22         | -1.76         | 8.85      | 10.98     | -11.63    | 3.66         | 18.28                |
|       |                        | 1341           | 1.19          | -1.20         | -1.71         | 8.57      | 10.68     | -11.29    | 3.55         | 17.74                |
|       |                        | 1906           | 1.16          | -1.19         | -1.68         | 8.34      | 10.43     | -11.01    | 3.46         | 17.30                |
|       |                        | $\infty$       | 0.56          | -1.79         | -1.72         | 7.27      | -16.53    | 7.40      | 3.90         | 19.51                |
|       | M06-2X/<br>def2-TZVPPD | 1064           | 1.21          | -1.12         | -1.75         | 9.23      | 10.45     | -11.89    | 3.67         | 18.33                |
|       |                        | 1341           | 1.18          | -1.10         | -1.71         | 8.95      | 10.18     | -11.56    | 3.56         | 17.82                |
|       |                        | 1906           | 1.16          | -1.08         | -1.68         | 8.73      | 9.96      | -11.29    | 3.48         | 17.40                |
|       |                        | $\infty$       | 0.55          | -1.96         | -1.54         | 7.79      | -16.47    | 7.95      | 3.97         | 19.87                |
|       | $\omega$ B97XD         | 1064           | 1.29          | -1.43         | -2.33         | 8.74      | 9.76      | -12.37    | 3.60         | 18.02                |
|       |                        | 1341           | 1.26          | -1.41         | -2.27         | 8.47      | 9.49      | -12.02    | 3.50         | 17.50                |
|       |                        | 1906           | 1.23          | -1.39         | -2.22         | 8.25      | 9.27      | -11.72    | 3.41         | 17.07                |
|       |                        | $\infty$       | 0.73          | -1.54         | -1.48         | 7.17      | -15.88    | 8.20      | 3.85         | 19.25                |

**Table S8:** Selected components contributing tensors and values of the second-harmonics generation,  $\beta_{\text{SHG}} (\beta^\lambda(-2\omega; \omega, \omega) \times 10^{-30} \text{ esu})$  of OPPh<sub>3</sub>py calculated at different levels of theory. All calculations employed the 6-311++G(2df,p) basis set unless otherwise indicated.

| Model | Method                 | $\lambda$ (nm) | $\beta_{xxx}$ | $\beta_{yyy}$ | $\beta_{zzz}$ | $\beta_x$ | $\beta_y$ | $\beta_z$ | $\beta_{  }$ | $\beta_{\text{SHG}}$ |
|-------|------------------------|----------------|---------------|---------------|---------------|-----------|-----------|-----------|--------------|----------------------|
| GAS   | B3LYP                  | 1064           | 0.57          | -0.51         | 2.97          | 5.05      | 8.92      | 7.67      | 2.56         | 12.80                |
|       |                        | 1341           | 0.55          | -0.49         | 1.93          | 4.86      | 8.28      | 4.57      | 2.13         | 10.64                |
|       |                        | 1906           | 0.53          | -0.47         | 1.36          | 4.65      | 7.75      | 2.92      | 1.90         | 9.50                 |
|       | CAM-B3LYP              | 1064           | 0.81          | -0.49         | -1.10         | 6.10      | 8.45      | -5.71     | 2.38         | 11.88                |
|       |                        | 1341           | 0.75          | -0.48         | -1.06         | 5.58      | 7.79      | -5.33     | 2.19         | 10.96                |
|       |                        | 1906           | 0.71          | -0.47         | -1.03         | 5.19      | 7.29      | -5.02     | 2.05         | 10.26                |
|       | M06-2X                 | 1064           | 0.71          | -0.42         | -0.65         | 5.86      | 8.70      | -4.98     | 2.32         | 11.61                |
|       |                        | 1341           | 0.66          | -0.41         | -0.66         | 5.34      | 8.01      | -4.68     | 2.14         | 10.71                |
|       |                        | 1906           | 0.62          | -0.40         | -0.65         | 4.95      | 7.50      | -4.42     | 2.00         | 10.01                |
|       | M06-2X/<br>def2-TZVPPD | 1064           | 0.63          | -0.39         | -0.55         | 5.84      | 7.92      | -4.79     | 2.19         | 10.94                |
|       |                        | 1341           | 0.58          | -0.37         | -0.58         | 5.37      | 7.36      | -4.57     | 2.04         | 10.19                |
|       |                        | 1906           | 0.55          | -0.36         | -0.59         | 5.01      | 6.93      | -4.36     | 1.92         | 9.60                 |
|       | $\omega$ B97XD         | 1064           | 0.70          | -0.49         | -1.03         | 5.51      | 8.38      | -5.81     | 2.32         | 11.59                |
|       |                        | 1341           | 0.65          | -0.48         | -0.99         | 5.01      | 7.74      | -5.40     | 2.14         | 10.69                |
|       |                        | 1906           | 0.61          | -0.47         | -0.95         | 4.64      | 7.27      | -5.07     | 2.00         | 10.00                |
| PCM   | B3LYP                  | 1064           | 0.97          | -1.00         | 1.61          | 8.20      | 12.55     | -0.45     | 3.00         | 14.99                |
|       |                        | 1341           | 0.92          | -0.97         | 0.85          | 7.57      | 11.43     | -2.23     | 2.78         | 13.89                |
|       |                        | 1906           | 0.87          | -0.94         | 0.43          | 7.06      | 10.58     | -3.09     | 2.62         | 13.09                |
|       | CAM-B3LYP              | 1064           | 1.24          | -0.95         | -1.99         | 9.06      | 11.55     | -11.74    | 3.76         | 18.80                |
|       |                        | 1341           | 1.14          | -0.92         | -1.85         | 8.20      | 10.55     | -10.73    | 3.43         | 17.13                |
|       |                        | 1906           | 1.07          | -0.89         | -1.74         | 7.56      | 9.81      | -9.95     | 3.18         | 15.89                |
|       | M06-2X                 | 1064           | 1.14          | -0.76         | -1.24         | 8.77      | 12.21     | -10.50    | 3.67         | 18.34                |
|       |                        | 1341           | 1.04          | -0.74         | -1.18         | 7.90      | 11.14     | -9.62     | 3.34         | 16.70                |
|       |                        | 1906           | 0.97          | -0.73         | -1.11         | 7.26      | 10.35     | -8.93     | 3.10         | 15.48                |
|       | M06-2X/<br>def2-TZVPPD | 1064           | 1.09          | -0.71         | -1.19         | 8.84      | 11.42     | -10.56    | 3.58         | 17.89                |
|       |                        | 1341           | 1.00          | -0.68         | -1.14         | 8.03      | 10.50     | -9.73     | 3.28         | 16.41                |
|       |                        | 1906           | 0.94          | -0.66         | -1.09         | 7.42      | 9.81      | -9.08     | 3.06         | 15.29                |
|       | $\omega$ B97XD         | 1064           | 1.21          | -0.94         | -1.86         | 8.65      | 11.52     | -11.69    | 3.71         | 18.56                |
|       |                        | 1341           | 1.11          | -0.91         | -1.73         | 7.82      | 10.55     | -10.67    | 3.38         | 16.92                |
|       |                        | 1906           | 1.04          | -0.89         | -1.62         | 7.20      | 9.83      | -9.90     | 3.14         | 15.70                |

**Table S8:** (Continue)

| Model | Method                 | $\lambda$ (nm) | $\beta_{xxx}$ | $\beta_{yyy}$ | $\beta_{zzz}$ | $\beta_x$ | $\beta_y$ | $\beta_z$ | $\beta_{  }$ | $\beta_{SHG}$ |
|-------|------------------------|----------------|---------------|---------------|---------------|-----------|-----------|-----------|--------------|---------------|
| SMD   | B3LYP                  | 1064           | 1.15          | -1.33         | 0.56          | 9.33      | 12.54     | -3.60     | 3.21         | 16.04         |
|       |                        | 1341           | 1.08          | -1.27         | -0.05         | 8.56      | 11.34     | -4.91     | 3.01         | 15.03         |
|       |                        | 1906           | 1.02          | -1.22         | -0.37         | 7.96      | 10.44     | -5.46     | 2.84         | 14.22         |
|       | CAM-B3LYP              | 1064           | 1.36          | -1.27         | -2.83         | 9.92      | 11.12     | -14.18    | 4.11         | 20.57         |
|       |                        | 1341           | 1.26          | -1.22         | -2.61         | 8.96      | 10.14     | -12.90    | 3.74         | 18.69         |
|       |                        | 1906           | 1.18          | -1.18         | -2.44         | 8.25      | 9.42      | -11.93    | 3.46         | 17.30         |
|       | M06-2X                 | 1064           | 1.29          | -1.10         | -1.99         | 9.79      | 11.99     | -12.89    | 4.03         | 20.14         |
|       |                        | 1341           | 1.18          | -1.07         | -1.85         | 8.80      | 10.92     | -11.73    | 3.66         | 18.29         |
|       |                        | 1906           | 1.10          | -1.03         | -1.73         | 8.07      | 10.13     | -10.86    | 3.38         | 16.90         |
|       | M06-2X/<br>def2-TZVPPD | 1064           | 1.24          | -1.03         | -1.94         | 9.98      | 11.23     | -12.99    | 3.97         | 19.86         |
|       |                        | 1341           | 1.15          | -0.98         | -1.81         | 9.05      | 10.29     | -11.89    | 3.63         | 18.14         |
|       |                        | 1906           | 1.07          | -0.94         | -1.71         | 8.36      | 9.60      | -11.05    | 3.37         | 16.85         |
|       | $\omega$ B97XD         | 1064           | 1.34          | -1.33         | -2.62         | 9.55      | 10.63     | -13.73    | 3.96         | 19.82         |
|       |                        | 1341           | 1.23          | -1.27         | -2.41         | 8.62      | 9.69      | -12.49    | 3.60         | 18.01         |
|       |                        | 1906           | 1.15          | -1.23         | -2.25         | 7.94      | 9.00      | -11.56    | 3.33         | 16.66         |

**Table S9:** Selected components contributing tensors and values of the static and frequency dependant second-order hyperpolarisability,  $\gamma$  (static  $\gamma(0; 0, 0, 0) \times 10^{-36}$  esu, dynamic  $\gamma^\lambda(-\omega; \omega, 0, 0) \times 10^{-36}$  esu) of OPPh<sub>3</sub>py calculated at different levels of theory. All calculations employed the 6-311++G(2df,p) basis set unless otherwise indicated.

| Model | Method                 | $\lambda$ | $\gamma_{xxyy}$ | $\gamma_{yyzz}$ | $\gamma_{xxzz}$ | $\gamma_{xxxx}$ | $\gamma_{yyyy}$ | $\gamma_{zzzz}$ | $\gamma$ |
|-------|------------------------|-----------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|----------|
| GAS   | B3LYP                  | 1064      | 14.94           | 18.82           | 15.71           | 47.98           | 40.15           | 106.44          | 58.62    |
|       |                        | 1341      | 14.49           | 18.09           | 15.17           | 46.62           | 39.10           | 100.50          | 56.30    |
|       |                        | 1906      | 14.13           | 17.52           | 14.74           | 45.51           | 38.24           | 95.91           | 54.47    |
|       |                        | $\infty$  | 17.32           | 23.42           | 11.15           | 44.18           | 58.96           | 57.04           | 52.79    |
|       | CAM-B3LYP              | 1064      | 11.22           | 12.86           | 11.54           | 35.75           | 30.77           | 60.81           | 39.66    |
|       |                        | 1341      | 10.96           | 12.52           | 11.26           | 34.94           | 30.12           | 58.82           | 38.64    |
|       |                        | 1906      | 10.74           | 12.24           | 11.03           | 34.27           | 29.58           | 57.21           | 37.80    |
|       |                        | $\infty$  | 12.81           | 15.36           | 8.45            | 33.47           | 41.52           | 36.80           | 37.01    |
|       | M06-2X                 | 1064      | 10.97           | 12.58           | 11.15           | 34.78           | 29.64           | 58.91           | 38.49    |
|       |                        | 1341      | 10.70           | 12.23           | 10.87           | 33.97           | 28.99           | 56.91           | 37.46    |
|       |                        | 1906      | 10.47           | 11.95           | 10.63           | 33.29           | 28.45           | 55.29           | 36.62    |
|       |                        | $\infty$  | 12.42           | 15.00           | 8.03            | 32.38           | 40.52           | 35.33           | 35.82    |
|       | M06-2X/<br>def2-TZVPPD | 1064      | 9.45            | 11.32           | 10.31           | 30.95           | 26.21           | 53.62           | 34.53    |
|       |                        | 1341      | 9.23            | 11.03           | 10.06           | 30.28           | 25.69           | 51.85           | 33.65    |
|       |                        | 1906      | 9.05            | 10.79           | 9.85            | 29.72           | 25.26           | 50.41           | 32.94    |
|       |                        | $\infty$  | 11.02           | 13.56           | 7.13            | 28.66           | 37.16           | 32.08           | 32.26    |
|       | $\omega$ B97XD         | 1064      | 10.84           | 12.23           | 10.96           | 34.27           | 30.12           | 58.08           | 38.06    |
|       |                        | 1341      | 10.60           | 11.92           | 10.71           | 33.53           | 29.51           | 56.28           | 37.13    |
|       |                        | 1906      | 10.39           | 11.66           | 10.50           | 32.92           | 29.00           | 54.82           | 36.36    |
|       |                        | $\infty$  | 12.29           | 14.69           | 8.28            | 32.26           | 39.75           | 35.64           | 35.63    |
| PCM   | B3LYP                  | 1064      | 25.36           | 28.38           | 23.80           | 78.63           | 69.76           | 149.35          | 90.89    |
|       |                        | 1341      | 24.58           | 27.32           | 23.02           | 76.38           | 67.88           | 141.78          | 87.56    |
|       |                        | 1906      | 23.95           | 26.48           | 22.39           | 74.54           | 66.34           | 135.86          | 84.91    |
|       |                        | $\infty$  | 32.60           | 38.72           | 21.11           | 85.73           | 104.13          | 103.23          | 95.59    |
|       | CAM-B3LYP              | 1064      | 18.93           | 19.77           | 17.80           | 58.07           | 52.65           | 88.56           | 62.70    |
|       |                        | 1341      | 18.46           | 19.22           | 17.35           | 56.72           | 51.49           | 85.73           | 61.08    |
|       |                        | 1906      | 18.08           | 18.78           | 16.98           | 55.60           | 50.53           | 83.43           | 59.75    |
|       |                        | $\infty$  | 24.00           | 26.60           | 15.92           | 63.90           | 73.93           | 67.46           | 67.67    |
|       | M06-2X                 | 1064      | 18.79           | 19.72           | 17.36           | 57.27           | 51.48           | 86.02           | 61.52    |
|       |                        | 1341      | 18.31           | 19.15           | 16.91           | 55.90           | 50.30           | 83.17           | 59.88    |
|       |                        | 1906      | 17.92           | 18.69           | 16.53           | 54.77           | 49.33           | 80.86           | 58.54    |
|       |                        | $\infty$  | 23.61           | 26.20           | 15.38           | 62.90           | 73.48           | 65.36           | 66.43    |
|       | M06-2X/<br>def2-TZVPPD | 1064      | 17.14           | 18.65           | 16.75           | 53.68           | 48.66           | 81.29           | 58.00    |
|       |                        | 1341      | 16.72           | 18.14           | 16.33           | 52.47           | 47.63           | 78.67           | 56.53    |
|       |                        | 1906      | 16.38           | 17.72           | 15.98           | 51.46           | 46.78           | 76.54           | 55.33    |
|       |                        | $\infty$  | 22.31           | 25.03           | 14.77           | 59.26           | 71.81           | 63.17           | 63.70    |
|       | $\omega$ B97XD         | 1064      | 18.60           | 19.46           | 17.41           | 57.25           | 52.72           | 85.09           | 61.46    |
|       |                        | 1341      | 18.16           | 18.95           | 16.99           | 55.99           | 51.61           | 82.48           | 59.94    |
|       |                        | 1906      | 17.79           | 18.53           | 16.64           | 54.94           | 50.69           | 80.34           | 58.70    |
|       |                        | $\infty$  | 23.64           | 25.86           | 15.96           | 63.95           | 73.47           | 65.56           | 66.78    |

**Table S9:** (Continue)

| Model | Method                 | $\lambda$ | $\gamma_{xxyy}$ | $\gamma_{yyzz}$ | $\gamma_{xxzz}$ | $\gamma_{xxxx}$ | $\gamma_{yyyy}$ | $\gamma_{zzzz}$ | $\gamma$ |
|-------|------------------------|-----------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|----------|
| SMD   | B3LYP                  | 1064      | 25.86           | 28.43           | 24.85           | 81.00           | 70.68           | 152.90          | 92.95    |
|       |                        | 1341      | 25.06           | 27.37           | 24.02           | 78.65           | 68.76           | 145.24          | 89.55    |
|       |                        | 1906      | 24.41           | 26.52           | 23.35           | 76.72           | 67.19           | 139.22          | 86.82    |
|       |                        | $\infty$  | 34.01           | 39.51           | 21.73           | 88.89           | 106.18          | 108.20          | 98.75    |
|       | CAM-B3LYP              | 1064      | 19.22           | 19.64           | 18.52           | 59.36           | 53.26           | 91.21           | 64.00    |
|       |                        | 1341      | 18.74           | 19.10           | 18.04           | 57.96           | 52.08           | 88.28           | 62.33    |
|       |                        | 1906      | 18.34           | 18.65           | 17.65           | 56.81           | 51.10           | 85.90           | 60.96    |
|       |                        | $\infty$  | 24.97           | 27.32           | 16.37           | 65.74           | 75.11           | 70.14           | 69.66    |
|       | M06-2X                 | 1064      | 19.20           | 19.66           | 18.18           | 58.78           | 51.94           | 88.93           | 63.01    |
|       |                        | 1341      | 18.70           | 19.09           | 17.70           | 57.36           | 50.75           | 85.98           | 61.31    |
|       |                        | 1906      | 18.29           | 18.62           | 17.30           | 56.18           | 49.77           | 83.58           | 59.92    |
|       |                        | $\infty$  | 24.77           | 26.96           | 15.97           | 64.92           | 75.14           | 68.43           | 68.78    |
|       | M06-2X/<br>def2-TZVPPD | 1064      | 17.62           | 18.74           | 17.63           | 55.22           | 49.40           | 84.00           | 59.64    |
|       |                        | 1341      | 17.18           | 18.22           | 17.17           | 53.96           | 48.35           | 81.28           | 58.11    |
|       |                        | 1906      | 16.82           | 17.80           | 16.80           | 52.91           | 47.47           | 79.07           | 56.86    |
|       |                        | $\infty$  | 23.62           | 25.92           | 15.49           | 61.38           | 74.20           | 66.39           | 66.41    |
|       | $\omega$ B97XD         | 1064      | 18.96           | 19.20           | 18.13           | 58.74           | 53.07           | 85.00           | 62.15    |
|       |                        | 1341      | 18.50           | 18.68           | 17.68           | 57.42           | 51.94           | 82.32           | 60.59    |
|       |                        | 1906      | 18.12           | 18.26           | 17.31           | 56.33           | 51.01           | 80.14           | 59.31    |
|       |                        | $\infty$  | 24.56           | 26.44           | 16.28           | 65.86           | 74.47           | 65.45           | 68.07    |



**Table S10:** Selected tensor components and field induced second-harmonic generation values,  $\gamma_{\text{ESHG}}(-2\omega; \omega, \omega, 0) \times 10^{-36}$  esu, of OPPh<sub>3</sub>py calculated at different levels of theory. All calculations employed the 6-311++G(2df,p) basis set unless otherwise indicated.

| Model | Method                 | $\lambda$ | $\gamma_{xxyy}$ | $\gamma_{yyzz}$ | $\gamma_{xxzz}$ | $\gamma_{xxxx}$ | $\gamma_{yyyy}$ | $\gamma_{zzzz}$ | $\gamma_{\text{ESHG}}$ |
|-------|------------------------|-----------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|------------------------|
| GAS   | B3LYP                  | 1064      | 17.99           | 19.48           | 24.13           | 56.87           | 46.99           | 157.38          | 76.66                  |
|       |                        | 1341      | 16.13           | 17.10           | 20.78           | 51.57           | 42.92           | 124.19          | 65.23                  |
|       |                        | 1906      | 14.85           | 15.57           | 18.66           | 47.74           | 39.96           | 105.37          | 58.20                  |
|       | CAM-B3LYP              | 1064      | 12.87           | 13.22           | 14.89           | 40.74           | 34.79           | 74.18           | 46.24                  |
|       |                        | 1341      | 11.90           | 12.21           | 13.67           | 37.83           | 32.45           | 66.14           | 42.35                  |
|       |                        | 1906      | 11.17           | 11.46           | 12.77           | 35.61           | 30.66           | 60.46           | 39.48                  |
|       | M06-2X                 | 1064      | 12.64           | 12.82           | 14.66           | 39.82           | 33.68           | 72.50           | 45.16                  |
|       |                        | 1341      | 11.65           | 11.81           | 13.41           | 36.87           | 31.32           | 64.29           | 41.20                  |
|       |                        | 1906      | 10.91           | 11.07           | 12.48           | 34.63           | 29.53           | 58.55           | 38.31                  |
|       | M06-2X/<br>def2-TZVPPD | 1064      | 10.74           | 11.75           | 13.08           | 35.10           | 29.40           | 65.73           | 40.18                  |
|       |                        | 1341      | 9.97            | 10.88           | 12.02           | 32.68           | 27.55           | 58.41           | 36.82                  |
|       |                        | 1906      | 9.39            | 10.23           | 11.24           | 30.83           | 26.12           | 53.31           | 34.37                  |
|       | $\omega$ B97XD         | 1064      | 12.34           | 12.48           | 14.06           | 38.81           | 33.84           | 70.06           | 44.01                  |
|       |                        | 1341      | 11.46           | 11.58           | 12.96           | 36.17           | 31.68           | 62.89           | 40.50                  |
|       |                        | 1906      | 10.79           | 10.90           | 12.14           | 34.14           | 30.01           | 57.76           | 37.90                  |
| PCM   | B3LYP                  | 1064      | 28.24           | 27.57           | 34.00           | 86.17           | 75.52           | 196.92          | 107.15                 |
|       |                        | 1341      | 25.36           | 24.59           | 29.84           | 78.15           | 68.86           | 160.73          | 93.14                  |
|       |                        | 1906      | 23.33           | 22.55           | 27.05           | 72.28           | 63.98           | 138.83          | 83.97                  |
|       | CAM-B3LYP              | 1064      | 20.27           | 19.55           | 22.04           | 61.53           | 55.27           | 101.01          | 68.05                  |
|       |                        | 1341      | 18.66           | 18.04           | 20.22           | 57.01           | 51.39           | 90.36           | 62.34                  |
|       |                        | 1906      | 17.46           | 16.91           | 18.87           | 53.56           | 48.43           | 82.76           | 58.11                  |
|       | M06-2X                 | 1064      | 20.20           | 19.10           | 22.05           | 60.85           | 54.24           | 98.75           | 67.07                  |
|       |                        | 1341      | 18.55           | 17.57           | 20.15           | 56.24           | 50.30           | 87.91           | 61.23                  |
|       |                        | 1906      | 17.32           | 16.44           | 18.75           | 52.74           | 47.30           | 80.24           | 56.93                  |
|       | M06-2X/<br>def2-TZVPPD | 1064      | 18.11           | 18.28           | 20.65           | 56.23           | 50.33           | 92.38           | 62.33                  |
|       |                        | 1341      | 16.73           | 16.88           | 18.96           | 52.20           | 46.94           | 82.45           | 57.15                  |
|       |                        | 1906      | 15.70           | 15.84           | 17.70           | 49.12           | 44.35           | 75.43           | 53.33                  |
|       | $\omega$ B97XD         | 1064      | 19.74           | 18.94           | 21.51           | 60.14           | 54.88           | 96.13           | 66.08                  |
|       |                        | 1341      | 18.25           | 17.56           | 19.83           | 55.94           | 51.22           | 86.41           | 60.80                  |
|       |                        | 1906      | 17.13           | 16.51           | 18.57           | 52.71           | 48.41           | 79.42           | 56.86                  |

**Table S10:** (Continue)

| Model | Method                 | $\lambda$ | $\gamma_{xxyy}$ | $\gamma_{yyzz}$ | $\gamma_{xxzz}$ | $\gamma_{xxxx}$ | $\gamma_{yyyy}$ | $\gamma_{zzzz}$ | $\gamma_{\text{ESHG}}$ |
|-------|------------------------|-----------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|------------------------|
| SMD   | B3LYP                  | 1064      | 28.67           | 28.84           | 34.15           | 88.59           | 76.14           | 199.57          | 108.95                 |
|       |                        | 1341      | 25.72           | 25.68           | 29.96           | 80.21           | 69.40           | 163.56          | 94.80                  |
|       |                        | 1906      | 23.63           | 23.51           | 27.14           | 74.10           | 64.46           | 141.57          | 85.48                  |
|       | CAM-B3LYP              | 1064      | 20.50           | 20.42           | 21.96           | 62.77           | 55.70           | 103.76          | 69.29                  |
|       |                        | 1341      | 18.86           | 18.79           | 20.13           | 58.09           | 51.76           | 92.78           | 63.42                  |
|       |                        | 1906      | 17.63           | 17.57           | 18.77           | 54.52           | 48.75           | 84.94           | 59.07                  |
|       | M06-2X                 | 1064      | 20.53           | 20.07           | 22.02           | 62.25           | 54.42           | 101.68          | 68.44                  |
|       |                        | 1341      | 18.83           | 18.42           | 20.11           | 57.46           | 50.45           | 90.51           | 62.42                  |
|       |                        | 1906      | 17.56           | 17.20           | 18.70           | 53.84           | 47.43           | 82.60           | 58.00                  |
|       | M06-2X/<br>def2-TZVPPD | 1064      | 18.50           | 19.31           | 20.79           | 57.56           | 50.74           | 94.98           | 63.77                  |
|       |                        | 1341      | 17.06           | 17.78           | 19.06           | 53.37           | 47.30           | 84.76           | 58.41                  |
|       |                        | 1906      | 15.99           | 16.64           | 17.78           | 50.18           | 44.67           | 77.52           | 54.47                  |
|       | $\omega$ B97XD         | 1064      | 20.05           | 19.79           | 21.28           | 61.59           | 55.06           | 96.09           | 66.72                  |
|       |                        | 1341      | 18.51           | 18.30           | 19.58           | 57.21           | 51.35           | 86.19           | 61.31                  |
|       |                        | 1906      | 17.35           | 17.17           | 18.31           | 53.86           | 48.49           | 79.07           | 57.27                  |

List of authors in **Reference 40**: Frisch M. J., Trucks G. W., Schlegel H. B., Scuseria G. E., Robb M. A., Cheeseman J. R., Scalmani G., Barone V., Petersson G. A., Nakatsuji H., Li X., Caricato M., Marenich A. V., Bloino J., Janesko B. G., Gomperts R., Mennucci B., Hratchian H. P., Ortiz J. V., Izmaylov A. F., Sonnenberg J. L., Williams-Young D., Ding F., Lipparini F., Egidi F., Goings J., Peng B., Petrone A., Henderson T., Ranasinghe D., Zakrzewski V. G., Gao J., Rega N., Zheng G., Liang W., Hada M., Ehara M., Toyota K., Fukuda R., Hasegawa J., Ishida M., Nakajima T., Honda Y., Kitao O., Nakai H., Vreven T., Throssell K., Montgomery Jr. J. A., Peralta J. E., Ogliaro F., Bearpark M. J., Heyd J. J., Brothers E. N., Kudin K. N., Staroverov V. N., Keith T. A., Kobayashi R., Normand J., Raghavachari K., Rendell A. P., Burant J. C., Iyengar S. S., Tomasi J., Cossi M., Millam J. M., Klene M., Adamo C., Cammi R., Ochterski J. W., Martin R. L., Morokuma K., Farkas O., Foresman J. B., Fox D. J. *Gaussian 16*, Revision C.01; Gaussian, Inc., Wallingford, CT, **2019**.

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